

Practical 3

* Aim:- Import a dataset with missing values, inspect the data, and handle missing values using techniques like deletion, mean imputation, and forward / backward fill.

* Theory:- Missing values are data entries that are not available or unknown. In Python, missing values are represented using NaN (Not a Number), which is provided by Numpy as `np.nan`.

* Missing values may occur because:-

- (i) Data was not entered
- (ii) Information was lost
- (iii) The value was not applicable.

* Why handle missing values?

Most data analysis and machine learning algorithms cannot work properly when missing values are present. Handling missing data improves:-

- (i) Accuracy of Analysis
- (ii) Data quality
- (iii) Reliability of results.

* Libraries Used:-

- (i) Pandas:- Pandas is used for creating and manipulating datasets in table format (DataFrame).
- (ii) Numpy:- Numpy provides `np.nan` to represent missing values.

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* Dataset description :- The program creates a student dataset with three columns :-

(i) Name - student name

(ii) Age - Student age

(iii) Marks - Students marks

Some values in Age & Marks are missing

* df.info() :- Displays no. of rows & columns, data types and Count of non-null values.

* df.isnull().sum() :- Shows the number of missing values in each column.

* Techniques used for handling missing values :-

(i) Deletion Method (dropna()) :- Removes all rows that contains any missing values.

(ii) Mean Imputation (fillna(mean)) :- Replaces missing numeric values with the average (mean) of the column.

(iii) Forward Fill (ffill) :- Replaces missing values with the previous valid value in the same column.

(iv) Backward Fill (bfill) :- Replaces missing values with the next valid value in the same column.

* Conclusion :- In this practical, missing values were identified and handled using deletion, mean imputation, forward fill and backward fill. These techniques helps clean the dataset and improve the accuracy and reliability of data analysis.

Practical 4

* Aim:- Use Python libraries like Pandas and Numpy to identify outliers using statistical methods and handle them by removing or capping.

* Theory:- Outliers are data values that are significantly different from the majority of observations in a dataset. These ^{un-}usual values may occur due to :-

- (i) Data entry errors
- (ii) Measurement errors
- (iii) Rare events
- (iv) Natural variation in data

For example, in a class where most students score between 60 and 90 marks, a value of 250 would be considered an outlier.

Outliers can:-

- (i) Distort the mean and standard deviation.
- (ii) Affect statistical analysis
- (iii) Reduce machine learning model accuracy
- (iv) Lead to misleading conclusions.

Therefore, detecting and handling outliers is an important part of data preprocessing.

* Libraries Used :-

- (i) Pandas :- Pandas is used to create and manipulate datasets in Data Frame format.
- (ii) Numpy :- Numpy is used for Numerical operations and calculations.

* Method for detecting Outliers :-

Interquartile Range (IQR) method :-

The IQR method identifies outliers based on the spread of the middle 50% of data.

$$IQR = Q_3 - Q_1$$

* Lower and upper limits :-

(i) Lower limit = $Q_1 - 1.5 \times IQR$

(ii) Upper limit = $Q_3 + 1.5 \times IQR$

Values outside these limits are considered outliers.

* Methods for removing or handling outliers :-

- (i) Removing outliers :- Rows containing outliers are deleted from dataset.
- (ii) Capping :- Extreme values are replaced with the lower or upper limit instead of removing them.

* Conclusion :- In this practical, outliers were identified using the IQR method and handled by removing or capping them. These techniques help reduce the impact of extreme values and improve the accuracy and reliability.